COMPLETE SOLVED PAPERS

Computer Science | Class 10

Order: Model Paper → Second Annual 2026 → First Annual 2026: B, D, M

1. Model Paper Solution

Computer Science SSC-II | FBISE Model Paper 2025 | Sections B and C

Complete solutions in simple English, including all OR options. Short answers follow the printed 2-mark questions; long answers cover every requested part. In the examination, follow the instructions for choosing between OR alternatives and write in the allotted spaces.

Section B - Q.2: Short Questions (22 marks)

(i) Define deadlock with an example. (1+1 marks)

Definition: A deadlock occurs when two or more processes keep waiting for resources held by one another, so none of them can continue.

Example: Process A holds a printer and waits for a scanner. Process B holds the scanner and waits for the printer. Neither process releases its resource, so both remain blocked.

OR - What is the role of data visualization in the data science process?

Data visualization presents data through charts, graphs, and maps. It helps people identify patterns, trends, relationships, and unusual values.

It also makes findings easier to explain, helping people make informed decisions.

(ii) A computer runs slowly when multiple applications are open. Which system-software functions are involved, and how can performance be improved? (1+1 marks)

Functions involved: The operating system's process management shares CPU time among programs, while memory management allocates RAM. Too many applications can overload these resources and cause slow performance.

Improvement: Close unnecessary applications and background processes, disable unnecessary startup programs, and increase RAM if memory is regularly insufficient.

OR - Which search algorithm is suitable for quickly retrieving customer data from a large database? How does it improve efficiency?

Binary search is suitable when the customer records are sorted by the search field, such as customer ID.

It checks the middle record and removes half the remaining search range at each step. This requires far fewer comparisons than checking every record individually.

(iii) Calculate the Mean Squared Error (MSE). (2 marks)

Actual values: [3, 5, 2, 8, 7]

Predicted values: [2.5, 5.3, 1.8, 7.5, 6.8]

Formula: MSE = sum of (Actual value - Predicted value)^2 / n

Actual value	Predicted value	Error	Squared error
3	2.5	0.5	0.25
5	5.3	-0.3	0.09
2	1.8	0.2	0.04
8	7.5	0.5	0.25
7	6.8	0.2	0.04
Total			0.67

MSE = 0.67 / 5

Answer: MSE = 0.134

OR - Convert $(2F08)_{16}$ into octal and binary. (1+1 marks)

Step 1: Hexadecimal to binary. Replace each hexadecimal digit with four binary digits.

Hexadecimal digit	Binary
2	0010
F	1111
0	0000
8	1000

Binary answer: $(2F08)_{16} = (0010\ 1111\ 0000\ 1000)_2$

Step 2: Binary to octal. Group the binary digits into sets of three from the right. Add leading zeros where needed:

000 010 111 100 001 000

The octal digits are 0, 2, 7, 4, 1, 0. Omit the unnecessary leading zero.

Octal answer: $(2F08)_{16} = (27410)_8$

(iv) Give a scenario in which all functionality is handled by the front end only. (2 marks)

A browser-based calculator can work entirely on the front end.

HTML creates its buttons and display, CSS styles them, and JavaScript performs the calculations in the browser. No back-end processing or database is required.

OR - Differentiate between system software and application software, with examples.

System software	Application software
Manages computer hardware and provides a platform for other programs.	Performs specific tasks for the user.
Examples: Windows and device drivers.	Examples: Microsoft Word and a web browser.

(v) Write four applications of a Real-Time Operating System. (2 marks)

A Real-Time Operating System (RTOS) is used where tasks must be completed within specified time limits.

1. Aircraft flight-control systems.
2. Vehicle braking and airbag control systems.
3. Industrial robot controllers.
4. Medical monitoring equipment.

OR - How does Blockchain technology provide security?

1. **Cryptographic linking:** Blocks are linked using hashes. Changing an old record changes its hash, making unauthorized changes detectable.
2. **Shared validation:** Network participants validate records according to agreed rules, making it difficult for one person to secretly change the transaction history.

(vi) How does fraud detection benefit from Machine Learning? (2 marks)

Machine Learning learns patterns from past transactions and identifies activities that may be fraudulent.

For example, it can flag an unusually large payment from an unfamiliar location. This helps organizations detect suspicious transactions quickly and reduce financial losses.

OR - Give two ways students can responsibly use social media when sharing information.

1. **Verify information:** Check facts using reliable sources before sharing, and credit the original source.
2. **Respect privacy and others:** Obtain permission before sharing someone's private information or photographs, and avoid harmful or insulting content.

(vii) List two characteristics of an algorithm. (1+1 marks)

1. **Definiteness:** Every step must be clear and unambiguous.
2. **Finiteness:** The algorithm must finish after a limited number of steps.

OR - Write two advantages of an online computing platform over an offline one.

1. **Remote access:** Users can access their work from different locations and devices through the internet.
2. **Collaboration:** Multiple users can work on and share the same project without repeatedly exchanging separate copies.

(viii) Why is breaking a program into smaller modules beneficial? Give two reasons. (2 marks)

1. **Easier understanding and maintenance:** Each module performs a specific task, making the program easier to read and update.
2. **Easier testing and debugging:** Modules can be tested separately, helping programmers locate and correct errors.

OR - How does choosing the right research method affect decision-making? Give an example. (1+1 marks)

A suitable research method provides relevant and reliable information. This reduces guesswork and supports better decisions.

Example: A school can survey students before choosing a new extracurricular activity. The survey reveals students' preferences and helps the school select a suitable activity.

(ix) Suggest two factors to consider when selecting a Machine Learning model. (2 marks)

1. **Nature of the task:** Select a model suitable for the problem, such as classification for detecting spam or regression for predicting prices.
2. **Available data:** Consider the amount, quality, and type of data, including whether labelled examples are available.

OR - How does network monitoring help prevent cyber threats? Give two justifications.

1. **Detects suspicious activity:** It can identify unusual traffic, repeated failed logins, or unauthorized connections.
2. **Supports quick action:** Alerts allow administrators to block suspicious connections or isolate affected devices before an attack spreads.

(x) How do shape, centre, and spread help explain a dataset? (2 marks)

- **Shape** shows the overall pattern, such as whether values are balanced or concentrated towards one side.
- **Centre** gives a typical value, such as the mean or median.
- **Spread** shows how close together or widely scattered the values are.

Example: Two classes may have the same average marks, but a larger spread means students' marks differ more widely.

OR - Why is email filtering important? Give two reasons.

1. It helps block phishing and fraudulent emails that try to steal passwords or financial information.
2. It helps identify spam and harmful attachments or links, reducing exposure to scams and malware.

(xi) How can AI be used responsibly in law enforcement? Give two suggestions. (2 marks)

1. **Ensure fairness and human review:** Check AI systems for bias and errors, and require trained people to review important decisions.
2. **Protect privacy:** Use only necessary, lawfully obtained data and protect it from unauthorized access.

OR - What is phishing, and how does it affect users? (1+1 marks)

Definition: Phishing is an attempt to trick users into revealing sensitive information or taking an unsafe action through fake messages or websites that appear trustworthy.

Effect: It can lead to stolen passwords, financial loss, identity theft, or unauthorized access to accounts.

Section C - Long Questions (20 marks)

Q.3 Explain the significance of an operating system. Describe file management and memory management. (1+4 marks)

Significance of an operating system - 1 mark

An operating system (OS) is system software that manages computer hardware and provides an environment in which applications run. It acts as an interface between users, applications, and hardware.

Examples: Windows, Linux, and macOS.

A. File management - 2 marks

File management organizes and controls files on storage devices. The operating system:

1. Allows users to create, open, save, rename, copy, move, and delete files.
2. Organizes files into folders and directories.
3. Keeps track of file locations and manages access permissions.

Example: When a student saves a document in a Computer Science folder, the operating system manages its storage and later retrieval.

B. Memory management - 2 marks

Memory management controls the use of main memory, or RAM. The operating system:

1. Keeps track of which memory locations are free or in use.
2. Allocates memory to programs when they need it.
3. Releases memory when it is no longer required and protects programs' memory areas from unauthorized access.

Example: When a browser and a word processor run together, the operating system allocates memory to both.

Q.3 OR - Discuss five components of digital literacy, with one example each. (5 marks)

Digital literacy is the ability to use digital technologies effectively, critically, safely, and responsibly.

Component	Explanation	Example
Information literacy	Finding information and checking its reliability and relevance.	Comparing reliable sources before using information in an assignment.
Digital communication and collaboration	Using digital tools to share ideas and work with others appropriately.	Working with classmates on a shared online document.
Digital content creation	Creating or editing digital material while respecting copyright.	Making a presentation with original text and properly credited images.
Digital safety and responsibility	Protecting information and behaving respectfully online.	Enabling two-factor authentication and obtaining permission before sharing a photograph.
Digital problem-solving	Selecting suitable digital tools and solving basic technical problems.	Using spreadsheet formulas to calculate students' average marks.

Q.4 Identify the use of a searching algorithm. Compare linear search and binary search. (1+4 marks)

Use - 1 mark: A searching algorithm finds a required item in a collection of data. It may report whether the item exists and give its position.

Example: Finding a student's record using a roll number.

Comparison - 4 marks

Feature	Linear search	Binary search
Method	Checks items one by one.	Checks the middle item and repeatedly halves the search range.
Data order	Works with sorted or unsorted data.	Requires sorted data.
Efficiency	May check many items in a large list.	Efficient for large sorted arrays because the search range shrinks quickly.
Worst-case comparisons	Up to n comparisons: O(n).	About log2(n) comparisons: O(log n), with direct access to middle items.

Example: Find 40 in [10, 20, 30, 40, 50, 60, 70].

Linear search: Checks 10, 20, 30, and then 40.

Binary search: Checks the middle value, 40, and finds it immediately.

Q.4 OR - Suggest five effective strategies to combat online hate speech. (5 marks)

1. **Educate users:** Teach people to recognize hate speech, understand its harm, and respect people from different backgrounds.
2. **Create and enforce clear rules:** Online communities should prohibit attacks against people based on characteristics such as religion, ethnicity, or gender.
3. **Report and moderate harmful content:** Users should report hate speech, and moderators should review reports and remove content that violates the rules.
4. **Avoid spreading or escalating abuse:** Do not repost hateful messages unnecessarily. Block abusive accounts and respond calmly where a constructive response is appropriate.
5. **Support targeted people:** Offer support, encourage safe reporting, and promote accurate information and respectful messages that challenge prejudice.

Q.5 How do IoT and Blockchain work together to improve cybersecurity? Give five suggestions. (5 marks)

The Internet of Things (IoT) connects devices that collect and exchange data. Blockchain can provide a shared record that makes unauthorized changes easier to detect.

1. **Verify device identities:** Register authorized device identities and use digital signatures to help verify which device sent a message.
2. **Protect record integrity:** Store important readings or their hashes on Blockchain so later changes to recorded information can be detected.
3. **Control access:** Use permission rules or smart contracts to specify which users and devices may submit records or perform particular actions.
4. **Maintain an audit trail:** Keep a time-ordered history of device readings and actions. This helps investigators trace suspicious activity.
5. **Reduce dependence on one record holder:** Maintain validated copies across multiple authorized participants, making it harder for one compromised record holder to secretly rewrite the history.

Example: Sensors monitor medicine temperatures during transport. A Blockchain-based record allows authorized staff to check readings and detect later changes.

Understanding: Devices still need secure passwords, updates, and protected communication. Blockchain cannot make a false sensor reading true.

Q.5 OR - Categorize HTML, CSS, JavaScript, Python, and PHP into front-end and back-end development. Give a reason for each. (5 marks)

Front-end development creates what users see and interact with in a browser. Back-end development handles processing and data on the server.

Technology	Category	Reason
HTML	Front end	Defines page structure and content, including headings, images, links, and forms.
CSS	Front end	Controls presentation, including colours, fonts, spacing, and layout.
JavaScript	Front end in this context	Adds browser interactivity, such as responding to clicks and checking form input.
Python	Back end	Can process requests, apply rules, and communicate with databases on a server.
PHP	Back end	Commonly handles server-side forms, sessions, and database access.

Example: HTML creates a login form, CSS styles it, and JavaScript checks whether fields are empty. A Python or PHP back end checks submitted login details.

Note: JavaScript can also be used on the back end, for example through Node.js.

Q.6 Write SQL queries for the given student table. (5 marks)

The paper does not specify a table name, so assume the table is named Students.

s_id	name	age	marks	grade	city
1	Saima	20	85	A	Lahore
2	Rida	22	78	B	Quetta
3	Minal	21	92	A	Karachi
4	Nida	23	65	C	Lahore
5	Seema	20	89	A	Quetta

(a) Retrieve all students who scored more than 80 marks.

```
SELECT *
FROM Students
WHERE marks > 80;
```

Explanation: WHERE selects only records with marks greater than 80.

Matching students: Saima, Minal, and Seema.

(b) Find the average marks of students.

```
SELECT AVG(marks) AS average_marks
FROM Students;
```

Explanation: AVG() calculates the mean of the marks.

Average = (85 + 78 + 92 + 65 + 89) / 5 = 409 / 5 = **81.8**.

(c) Get the names of students from Quetta or Lahore.

```
SELECT name
FROM Students
WHERE city = 'Quetta' OR city = 'Lahore';
```

Explanation: OR selects a student if either city condition is true.

Matching names: Saima, Rida, Nida, and Seema.

(d) Count the number of students in each grade.

```
SELECT grade, COUNT(*) AS student_count
FROM Students
GROUP BY grade;
```

Explanation: GROUP BY creates one group for each grade, and COUNT(*) counts its students.

Grade	Student count
A	3
B	1
C	1

(e) Retrieve students sorted by marks in descending order.

```
SELECT *
FROM Students
ORDER BY marks DESC;
```

Explanation: DESC places the highest marks first.

Position	Student	Marks
1	Minal	92
2	Seema	89
3	Saima	85
4	Rida	78
5	Nida	65

Q.6 OR - TechNova Solutions business case. (3+2 marks)

TechNova expects revenue growth of 50% per year and expense growth of 30% per year.

(a) Three strategies to maintain revenue growth if customer acquisition slows down - 3 marks

1. **Retain existing customers:** Improve support, fix problems promptly, and provide useful features so fewer customers cancel subscriptions.

2. **Increase revenue from existing customers:** Offer valuable premium plans, additional features, or extra services that existing customers are willing to purchase.
3. **Reach additional markets:** Adapt the product for new customer groups and use partnerships or referral programmes to reach potential buyers.

(b) Two financial risks if expenses grow faster than 30% per year - 2 marks

1. **Lower profit or losses:** Higher expenses reduce profit if revenue does not increase enough to cover them. Expenses may eventually exceed revenue.
2. **Cash-flow problems:** The company may use available cash more quickly and struggle to pay salaries, suppliers, or other bills. It may need additional borrowing or investment.

2. Second Annual Examination 2026

Computer Science SSC-II | Federal Board | Complete solution with OR options

Section B - Q.2: Short Questions (22 marks)

(i) What is the hexadecimal number system? State one use in computing. (2 marks)

The hexadecimal number system has base 16. It uses digits 0-9 and letters A-F, where A to F represent values 10 to 15.

Use: It represents colours in web development, such as #FF0000 for red.

OR - Define supervised learning and give one real-world application.

Supervised learning is a Machine Learning method in which a model learns from labelled examples containing inputs and their correct outputs.

Application: A spam filter learns from emails labelled "spam" and "not spam" to classify new emails.

(ii) A computer becomes slow when several programs run simultaneously. Identify one system-software function that can help and explain it. (2 marks)

Function: Process management. The operating system schedules CPU time among running programs and manages their execution.

It can give suitable priority to important tasks and suspend unnecessary processes, helping available resources be used more effectively.

OR - Which searching algorithm is suitable for finding a roll number in 5,000 sorted roll numbers? Give a reason.

Binary search is suitable because the roll numbers are already sorted.

It compares the required number with the middle number and removes half the remaining search range each time, so it is much faster than checking every number.

(iii) A bakery wants to predict daily sales. Identify two important data-science stages after collecting data. (2 marks)

1. **Data cleaning and preparation:** Correct mistakes, remove duplicate records, and handle missing values in the sales data.
2. **Model building:** Use the prepared data to train a model that predicts daily sales.

OR - Convert $(3A)_{16}$ into binary and octal.

Hexadecimal to binary: Each hexadecimal digit represents four binary digits.

3 = 0011; A = 1010.

Binary answer: $(3A)_{16} = (00111010)_2$

Binary to octal: Group digits into threes from the right, adding a leading zero where necessary:

000 111 010 = 0, 7, 2.

Octal answer: $(3A)_{16} = (72)_8$

(iv) Differentiate between front-end and back-end development. (2 marks)

Front-end development	Back-end development
Builds the part of a website that users see and interact with.	Handles server-side processing and database operations.
Common technologies: HTML, CSS, and JavaScript.	Common technologies: Python and PHP.

OR - Explain one role of system software in controlling interaction between hardware and applications.

System software acts as an interface between applications and hardware.

Example: When an application requests printing, the operating system and printer driver communicate with the printer and manage the printing operation.

(v) Write four applications of a Real-Time Operating System (RTOS). (2 marks)

An RTOS is used where tasks must be completed within specified time limits.

1. Vehicle braking and airbag control systems.
2. Industrial robot controllers.
3. Medical monitoring equipment.
4. Aircraft flight-control systems.

OR - Explain how cryptocurrency relies on Blockchain to record transactions.

Cryptocurrency transactions are checked according to network rules and recorded in blocks.

These blocks are linked using cryptographic hashes and shared across the network, creating a verifiable transaction history and helping prevent double spending.

(vi) Differentiate between supervised and unsupervised learning models. (2 marks)

Supervised learning	Unsupervised learning
Uses labelled data with known correct outputs.	Uses unlabelled data without supplied correct outputs.
Learns to predict a result, such as a house price.	Finds patterns or groups, such as groups of similar customers.

OR - State two ways an infographic or multimedia presentation can communicate information effectively.

1. Use suitable charts, images, and diagrams to explain information visually.
2. Present information in a clear order with short text and readable labels, supported by relevant audio or animation where useful.

(vii) Enlist two searching or sorting algorithms. (2 marks)

1. Binary search.
2. Bubble sort.

OR - Name two software tools that assist programmers in developing or managing projects.

1. **Microsoft Visual Studio:** An IDE used to write, run, and debug programs.
2. **Git:** A version-control tool used to track and manage code changes.

(viii) Why is dividing a large program into smaller modules or functions beneficial? Give two reasons. (2 marks)

1. **Easier understanding and maintenance:** Each module performs a specific task, making the program easier to read and update.
2. **Easier testing and debugging:** A faulty module can be tested and corrected separately.

OR - Name two digital tools used to collect customer or market information.

1. **Google Forms:** Collects information through online surveys.
2. **Google Trends:** Shows interest in search topics over time and across regions.

(ix) Which visualization is suitable for comparing the number of students in different classes? Give a reason. (2 marks)

A bar chart is suitable.

Each bar represents a class, and its height represents the number of students. Comparing bar heights makes differences between classes easy to see.

OR - State two strategies to prevent cyberbullying or online harassment.

1. Use privacy controls and limit unwanted contact from unknown users.
2. Promote respectful communication and promptly block and report abusive accounts or content.

(x) Define churn prediction and describe its role in customer retention. (2 marks)

Churn prediction uses customer data to identify people who are likely to stop using a company's service or product.

It helps the company retain customers by solving their problems, improving service, or offering suitable benefits before they leave.

OR - Define a deepfake video and explain two potential harms.

A deepfake video is a video created or altered using AI to make someone appear to say or do something they did not actually say or do.

1. **Damage to reputation:** False videos can embarrass or wrongly accuse a person.
2. **Fraud and misinformation:** They can impersonate trusted people, deceive viewers, or spread false claims.

(xi) Describe the role of AI in solving real-world problems. (2 marks)

AI analyses data, identifies patterns, and makes predictions or recommendations that help solve problems.

Examples: It can help doctors identify disease in medical images and help transport systems predict traffic conditions.

OR - Write two benefits of cloud computing.

1. **Remote access:** Users can access stored data and applications through an internet connection from different locations.
2. **Lower initial cost and flexibility:** Users can rent computing resources and increase or decrease them according to their needs.

Section C - Long Questions (20 marks)

Q.3 Explain the role of an operating system as a resource manager. Describe process management and device management. (5 marks)

An operating system manages computer resources, including the CPU, memory, storage, and input/output devices. It allocates these resources so programs can run correctly.

A. Process management

A process is a program that is currently running. The operating system:

1. Creates and ends processes.
2. Schedules CPU time among processes.
3. Coordinates their execution so several programs can run without unnecessary interference.

Example: While a user types a document and listens to music, the operating system shares CPU time between the programs.

B. Device management

The operating system:

1. Communicates with devices through device drivers.
2. Coordinates input and output operations.
3. Manages access to shared devices.

Example: If several documents are sent to one printer, the operating system places them in a print queue and sends them in order.

Q.3 OR - Explain how key ideas can be identified and extracted from given data. (5 marks)

1. **Identify the purpose:** Decide what question the data should answer.
2. **Review the data:** Examine headings, labels, values, text, and other information to understand its contents.
3. **Find important patterns:** Look for repeated ideas, trends, similarities, differences, and unusual values.
4. **Select relevant information:** Keep information connected to the question and leave out unrelated details.
5. **Summarize and check:** Express the main findings in simple words and verify them against the original data.

Example: In monthly sales data, the key ideas may be the overall sales trend, the best-selling product, and the month with the lowest sales.

Q.4 A teacher wants to arrange students' marks in ascending order. Identify an algorithm, explain its working with an example, and justify your choice. (5 marks)

Suitable algorithm: Insertion sort.

Insertion sort builds a sorted part of the list. It takes each new value and inserts it into the correct position among the values already sorted.

Example: Arrange [65, 40, 80, 55] in ascending order.

Stage	Action	List
Initial	Treat 65 as the first sorted value.	65, 40, 80, 55
Pass 1	Insert 40 before 65.	40, 65, 80, 55
Pass 2	80 is already after 65.	40, 65, 80, 55
Pass 3	Insert 55 between 40 and 65.	40, 55, 65, 80

Sorted marks: [40, 55, 65, 80]

Justification: Insertion sort is simple, requires little extra memory, and works well for small or nearly sorted lists, such as a classroom list of marks.

Q.4 OR - Suggest five strategies to prevent cyberbullying and online harassment. (5 marks)

1. **Teach responsible online behaviour:** Explain how hurtful posts affect others and encourage respectful communication.
2. **Use privacy settings:** Limit who can view profiles, send messages, or comment on posts.
3. **Block and report abuse:** Use the platform's reporting tools to report abusive accounts and content.
4. **Create clear rules and moderate communities:** Schools and online groups should define unacceptable behaviour and apply rules fairly.
5. **Provide a safe reporting and support system:** Encourage victims to tell a trusted adult or responsible authority, save evidence, and receive help without being blamed.

Q.5 Explain Blockchain's role in securing IoT systems. Describe six measures that reduce cyber threats. (2+3 marks)

Role of Blockchain in IoT security - 2 marks

Blockchain can store a shared, tamper-evident record of device identities, readings, and actions.

1. **Data integrity:** Cryptographically linked records make unauthorized changes detectable.
2. **Traceability:** Authorized users can check which device submitted a record and review its history.

Example: A transport company can record readings from temperature sensors to track the conditions of medicines during delivery.

Six measures to reduce cyber threats - 3 marks

1. **Change default passwords:** Use strong, unique passwords for devices and accounts.
2. **Update software and firmware:** Install security fixes for known weaknesses.
3. **Encrypt communication:** Protect data while it travels between devices and servers.
4. **Use firewalls:** Block unnecessary or unauthorized network connections.
5. **Control access:** Allow only authorized users and devices to access the system.
6. **Separate IoT devices from sensitive systems:** Use a separate network or network segment to limit the spread of an attack.

Q.5 OR - Differentiate front-end and back-end development. Explain HTML, CSS, JavaScript, PHP, and Python. (5 marks)

Front-end development creates the website interface that users see and interact with.

Back-end development handles server-side processing, business rules, and database access.

Technology	Role in web development
HTML	Defines page structure and content, including headings, links, images, and forms.
CSS	Controls appearance, including colours, fonts, spacing, and layout.
JavaScript	Adds interactivity, such as responding to clicks, checking input, and updating content.
PHP	Commonly performs server-side processing, such as handling forms, sessions, and database operations.
Python	Can build server-side applications that process requests, apply rules, and communicate with databases.

Example: On a registration page, HTML creates the form, CSS styles it, and JavaScript checks input. A PHP or Python back end checks the information and stores it in a database.

Q.6 Analyse participation in extracurricular activities. (5 marks)

Activity	Participants
Sports	120
Debates	45
Science Club	70
Arts	55
Computing Club	90

(a) Which visualization is most suitable?

A bar chart is most suitable. Place activities on the horizontal axis and the number of participants on the vertical axis.

(b) Why is this visualization appropriate?

The activities are separate categories. A bar chart makes their participant numbers easy to compare through the heights of the bars.

(c) Which activity has the highest participation?

Sports, with 120 participants.

(d) Which activity has the lowest participation?

Debates, with 45 participants.

(e) How can this visualization help the school make decisions?

It helps the school distribute funds, equipment, rooms, and staff according to participation. It also identifies activities, such as debates, that may need promotion or further investigation.

Q.6 OR - Pitch a digital business idea using the five given points. (5 marks)

Business idea: LearnEasy Online Tutoring

(a) Problem or need: Many Class 9 and 10 students need clear explanations and examination practice but cannot easily travel to tuition centres.

(b) Target customers: Class 9 and 10 students preparing for board examinations, especially those who need affordable support from home.

(c) Digital product or service: An online tutoring service offering live lessons, recorded explanations, practice tests, and question-answer sessions.

(d) Method of generating revenue: The business will charge a monthly subscription or a fee for each subject course.

(e) Digital promotion method: The business will publish free sample lessons on YouTube, with a link through which interested students can register for paid classes.

3. First Annual Examination 2026 - Paper B

Computer Science SSC-II | Federal Board | Complete solution with OR options

Section B - Q.2: Short Questions (22 marks)

(i) What is the binary system? Why is it used by computers? (1+1 marks)

The binary number system has base 2 and uses only two digits: 0 and 1.

Computers use binary because their electronic circuits have two basic states, such as OFF and ON, which can represent 0 and 1.

OR - Enlist any two purposes of data science.

1. **Finding patterns:** Data science finds useful patterns and relationships in data.
2. **Making predictions:** It uses past data to predict future results, such as sales or weather.

(ii) How does system software manage communication between hardware components? Give two points. (1+1 marks)

1. Device drivers allow the operating system to communicate with devices such as keyboards, printers, and storage drives.
2. The operating system coordinates data transfer between the CPU, memory, and input/output devices so they work together correctly.

OR - Why is understanding a real-world business problem important in data science?

1. It helps identify what problem must be solved, such as falling sales or customers leaving.
2. It helps select the relevant data and suitable methods, so the results are useful for the business.

(iii) What is a searching algorithm? Give one example. (1+1 marks)

A searching algorithm is a step-by-step method used to find a particular item in a collection of data.

Example: Linear search checks items one by one until the required item is found or the list ends.

OR - What is meant by signed and unsigned numbers?

- **Signed numbers** can represent negative numbers, zero, and positive numbers. Examples: -8, 0, and +8.
- **Unsigned numbers** represent only zero and positive numbers. Examples: 0, 8, and 25.

(iv) Name two technologies used in front-end development. (1+1 marks)

1. **HTML:** Defines the structure and content of a web page.
2. **CSS:** Controls appearance, including colours, fonts, and layout.

OR - Why should different audiences be considered while sharing information? Give two points.

1. Different audiences have different knowledge levels, so language and detail should match their understanding.
2. Different audiences have different needs and interests, so relevant examples and suitable formats should be used.

(v) What is meant by system software? (2 marks)

System software is a set of programs that manages computer hardware and provides a platform for application software to run.

Examples: Operating systems, device drivers, and utility programs.

OR - What is meant by the Internet of Things (IoT)?

The Internet of Things (IoT) is a network of physical devices connected to the internet that collect and exchange data.

Example: A smart temperature sensor sends room-temperature readings to a mobile application.

(vi) Briefly explain the scope of data science. (2 marks)

Data science includes collecting, cleaning, analysing, and interpreting data to find useful information and make predictions.

It is used in many fields, such as healthcare, education, banking, agriculture, and business.

OR - How does system software control input devices?

The operating system uses device drivers to communicate with input devices such as keyboards and mice.

It receives their input and makes it available to the correct application. For example, it sends keyboard input to a word processor.

(vii) What are the necessary conditions for binary search? Give two. (1+1 marks)

1. The data must be sorted, either in ascending or descending order.
2. Items must be comparable, so the algorithm can decide whether the required value lies before or after the middle item.

In an array, direct access to the middle item makes binary search efficient.

OR - What is meant by a code repository?

A code repository is a place where a project's source code and related files are stored and managed.

When used with version control, it also keeps a history of changes and helps programmers work together.

(viii) Why is abstraction important in program development? Give two reasons. (1+1 marks)

1. It reduces complexity by hiding unnecessary details and showing only important features.
2. It helps programmers create reusable solutions that can be used in different parts of a program.

OR - Why is digital research important for a business idea? Give two reasons.

1. It helps discover customer needs and preferences, so the business can offer a useful product.
2. It provides information about competitors, prices, and market trends, helping the business make better plans.

(ix) Name any two types of data visualizations. (1+1 marks)

1. Bar chart.
2. Line graph.

OR - State two negative impacts of cyberbullying on individuals.

1. It can cause stress, fear, anxiety, or sadness.
2. It can reduce confidence and concentration, affecting studies, work, and relationships.

(x) How does churn prediction help a business? (2 marks)

Churn prediction identifies customers who are likely to stop using a company's product or service.

The business can contact these customers, solve their problems, or offer suitable benefits to encourage them to stay.

OR - Why are AI applications important in daily life? Write two uses.

1. Navigation applications use AI to suggest suitable routes based on traffic.
2. Email services use AI to identify spam and move unwanted messages to the spam folder.

(xi) Give two examples where AI affects decision-making in daily life. (2 marks)

1. **Online shopping:** AI recommends products based on a customer's searches and previous purchases.
2. **Travel:** AI-based navigation recommends routes by analysing traffic conditions and estimated journey times.

OR - Why is information security important on the internet? Give two reasons.

1. It protects personal information, such as passwords and bank details, from unauthorized access.
2. It helps prevent fraud, identity theft, and misuse of information.

Section C - Long Questions (20 marks)

Q.3 Describe the role of the following software development tools, with one example each. (5 marks)

Tool	Role	Example
Translator	Converts source code into another form, such as machine code, so it can be executed.	GCC compiler
Integrated Development Environment (IDE)	Provides tools for writing, running, and debugging programs in one place.	Microsoft Visual Studio
Code repository	Stores project code and, with version control, keeps its change history.	A Git repository hosted on GitHub
Debugger	Helps find errors by running a program step by step and checking variable values.	GDB
Language editor	Allows programmers to write and edit source code, often with syntax highlighting.	Notepad++

Q.3 OR - How does data visualization help analysts understand patterns, trends, and relationships in a large dataset? Give five explained points. (5 marks)

1. **Summarizes large amounts of data:** Charts turn many rows of figures into a clear picture, making the main information easier to understand.
2. **Shows patterns:** Repeated behaviour becomes visible. For example, a sales chart may show that sales increase every weekend.
3. **Shows trends over time:** A line graph can show whether sales, costs, or profits are increasing or decreasing.
4. **Shows relationships:** A scatter plot can help show whether two quantities are related, such as advertising spending and sales.
5. **Highlights unusual values:** A chart can reveal an unusually high or low value. Analysts can investigate whether it represents an error or an important event.

Q.4 Make a trace table to sort [3, 1, 4, 1, 5, 9, 2, 6]. (5 marks)

Algorithm: Bubble sort in ascending order.

Bubble sort compares neighbouring values and swaps them if the left value is larger. In each pass, the largest remaining unsorted value moves towards the right.

Stage	List
Initial	3, 1, 4, 1, 5, 9, 2, 6
Pass 1	1, 3, 1, 4, 5, 2, 6, 9
Pass 2	1, 1, 3, 4, 2, 5, 6, 9
Pass 3	1, 1, 3, 2, 4, 5, 6, 9
Pass 4	1, 1, 2, 3, 4, 5, 6, 9
Pass 5	1, 1, 2, 3, 4, 5, 6, 9 - no swaps

Since no swaps occur in Pass 5, an optimized bubble sort stops.

Sorted list: [1, 1, 2, 3, 4, 5, 6, 9]

Q.4 OR - Describe five software methods used to protect devices. (5 marks)

1. **Antivirus software:** Detects and removes harmful programs such as viruses.
2. **Firewall:** Filters network traffic and blocks unauthorized connections.
3. **Software updates:** Fix security weaknesses in operating systems and applications.
4. **Access control:** Passwords and user permissions prevent unauthorized users from accessing a device or its files.
5. **Encryption:** Converts readable data into an unreadable form so unauthorized people cannot understand it without the correct key.

Q.5 How does IoT work? Describe sensors, connectivity, and data processing. (2+3 marks)

The Internet of Things connects physical devices so they can collect data, share it, and respond to conditions.

1. **Sensors collect data:** Sensors measure conditions such as temperature, movement, light, or moisture.
2. **Connectivity transfers data:** Devices send collected data through connections such as Wi-Fi, Bluetooth, or mobile networks.
3. **Data processing produces useful results:** A local computer or cloud system analyses the data and decides what action is needed.

Example: In a smart irrigation system, a soil-moisture sensor measures moisture. The reading is sent to a controller. If the soil is too dry, the controller turns on the water pump.

Q.5 OR - Why are front-end and back-end development essential for a dynamic website? Give examples. (2.5+2.5 marks)

Front-end development

The front end is the part of a website that users see and interact with. It includes text, images, menus, forms, and buttons. HTML, CSS, and JavaScript are commonly used to build it.

Example: An online shop's product page and Add to Cart button are front-end features.

Back-end development

The back end runs on the server. It processes requests, applies rules, and manages information stored in a database. Languages such as Python and PHP can be used for back-end development.

Example: The back end checks login details, retrieves product prices, and saves orders.

Working together: When a customer places an order, the front end sends order details to the back end. The back end saves the order and returns a confirmation.

Q.6 Define digital communication. List four digital tools used to communicate ideas to different audiences. (1+4 marks)

Digital communication is the exchange of information and ideas through electronic devices and digital networks.

1. **Email:** Used to send formal messages, notices, and documents.
2. **Presentation software:** PowerPoint or Google Slides presents ideas using text, images, and charts.
3. **Video-conferencing tools:** Zoom or Google Meet allows live discussions and online lessons.
4. **Social media platforms:** Used to share announcements, images, and videos with a wider audience.

Q.6 OR - Prepare a short business pitch for an online tutoring service, highlighting the problem, solution, and target customers. (5 marks)

Business name: LearnEasy Online

Problem: Many students find difficult subjects hard to understand and cannot easily travel to tuition centres.

Solution: LearnEasy Online provides affordable live lessons, recorded explanations, practice questions, and opportunities to ask teachers questions from home.

Target customers: Our main customers are students in Classes 9 and 10 who need help preparing for board examinations.

Main benefit: Students receive clear explanations and flexible study support while saving travel time and cost.

4. First Annual Examination 2026 - Paper D

Computer Science SSC-II | Federal Board | Complete solution with OR options

Section B - Q.2: Short Questions (22 marks)

(i) What is an encoding scheme? Write its role in data representation. (1+1 marks)

An encoding scheme is a set of rules that represents characters or other information using codes.

It allows computers to store, process, and exchange information in binary form. ASCII and Unicode are examples.

OR - What is Artificial Intelligence? Give one example.

Artificial Intelligence (AI) is the ability of a computer system to perform tasks that normally require human intelligence, such as learning, understanding language, and making decisions.

Example: An email spam filter identifies unwanted messages.

(ii) How does system software handle data storage operations? (2 marks)

System software manages how data is saved, organized, retrieved, and deleted on storage devices.

The operating system uses a file system to organize files and folders and device drivers to communicate with storage hardware.

OR - What is a sorting algorithm? Why is sorting important?

A sorting algorithm arranges data in a particular order, such as ascending or descending order.

Sorting makes data easier to understand and compare, and it allows efficient searching methods such as binary search to be used.

(iii) What is meant by building a model in the data science life cycle? (2 marks)

Building a model means using a suitable algorithm and prepared data to learn patterns.

The trained model can then make predictions or classify new data. For example, a model can use past sales to predict future sales.

OR - How does the binary number system differ from the octal number system?

Binary	Octal
Has base 2.	Has base 8.
Uses digits 0 and 1.	Uses digits 0 to 7.

Example: Decimal 8 is written as $(1000)_2$ in binary and $(10)_8$ in octal.

(iv) Name two technologies used in back-end development. (1+1 marks)

1. Python.
2. PHP.

Both can be used to process requests and work with databases on a web server.

OR - Briefly explain the role of system software in output operations.

System software uses device drivers to communicate with output devices such as monitors and printers.

It sends output data in a form the device can use and manages operations such as displaying information or printing documents.

(v) Give two examples of system software. (1+1 marks)

1. Windows operating system.
2. Printer device driver.

OR - What is Blockchain technology? Write one application.

Blockchain is a shared digital record in which data is stored in blocks linked using cryptographic hashes. Network participants follow agreed rules to validate new records.

Application: Recording cryptocurrency transactions, such as Bitcoin transfers.

(vi) Why is Machine Learning important in modern technology? Give two reasons. (2 marks)

1. It allows systems to learn patterns from data and make predictions without a separate rule being written for every situation.
2. It helps automate complex tasks, such as detecting spam, recognizing images, and recommending products.

OR - Give two examples of using digital tools to publish ideas.

1. Writing and publishing an article on a blog.
2. Uploading an educational video to YouTube.

(vii) Briefly explain inclusion and exclusion principles in counting problems. (2 marks)

The inclusion-exclusion principle counts items in overlapping groups without counting the same item twice.

First, add the numbers in the groups. Then subtract the number common to both:

n(A union B) = n(A) + n(B) - n(A intersection B)

Example: If 20 students like cricket, 15 like football, and 5 like both, the number who like at least one is:

20 + 15 - 5 = **30 students**.

OR - What is a translator in computer science? Give one example.

A translator converts a program written in one language into another form, such as machine code.

Example: A C++ compiler translates C++ source code into machine code.

(viii) Briefly explain modularity in software development. (2 marks)

Modularity means dividing a large program into smaller parts called modules or functions. Each part performs a particular task.

This makes the program easier to understand, test, update, and maintain.

OR - Name two digital tools used to collect customer feedback.

1. Google Forms.
2. SurveyMonkey.

(ix) How does data visualization help in decision-making? (2 marks)

Data visualization presents information through charts and graphs, making comparisons, patterns, and trends easier to see.

This helps people understand a situation quickly and make decisions based on evidence.

OR - State two hardware methods used to protect devices.

1. Use a UPS or surge protector to protect against power failures or voltage changes.
2. Use physical locks or cable locks to reduce theft and unauthorized physical access.

(x) How does behavioural segmentation affect business strategy? Give two points. (1+1 marks)

Behavioural segmentation divides customers into groups according to their actions, such as buying habits, product use, and loyalty.

1. It helps businesses send relevant offers to different customer groups.
2. It helps plan loyalty rewards and services based on how customers use the product.

OR - How is AI used in Natural Language Processing? Give two points.

1. AI helps computers understand human language, such as identifying the meaning or purpose of a message.
2. AI helps computers produce or transform language, such as answering questions, translating text, and summarizing documents.

(xi) How can AI affect employment in society? Give two points. (1+1 marks)

1. AI can automate some routine tasks, so certain jobs may decrease or change.
2. AI can create new jobs in areas such as AI development, data analysis, and system maintenance, increasing the need for new skills.

OR - State two practices that help protect personal data online.

1. Use strong, unique passwords and enable two-factor authentication.
2. Share personal information only when necessary and check the privacy settings of online accounts.

Section C - Long Questions (20 marks)

Q.3 Explain the role of an IDE. Describe at least four features with examples. (1+4 marks)

An Integrated Development Environment (IDE) is software that combines programming tools in one place. It helps programmers write, run, test, and debug programs.

Example: Microsoft Visual Studio.

1. **Code editor:** Allows programmers to write and edit code. Syntax highlighting uses different colours for keywords, strings, and other parts.
2. **Compiler or interpreter integration:** Allows the programmer to translate or execute the program from within the IDE.
3. **Debugger:** Helps locate errors by running the program step by step and checking variable values.
4. **Code completion:** Suggests variable names, function names, or keywords while the programmer types, saving time and reducing typing mistakes.

Q.3 OR - How are key ideas extracted in digital literacy? Write five steps. (5 marks)

1. **Identify the purpose:** Decide what information you need and why you need it.
2. **Review the material:** Read the headings, introduction, main text, and conclusion, or watch the material carefully.
3. **Identify important points:** Look for repeated ideas, keywords, major facts, and main arguments.
4. **Separate main ideas from details:** Keep information that answers the main question and remove unrelated examples or repetition.
5. **Summarize in your own words:** Write a short, clear summary and check that it correctly represents the original material.

Q.4 Differentiate between searching and sorting algorithms. Give one example each. (2.5+2.5 marks)

Feature	Searching algorithm	Sorting algorithm
Purpose	Finds a required item in data.	Arranges data in a particular order.
Result	Shows whether an item exists and may give its position.	Produces an ordered collection of data.
Order requirement	Some methods need sorted data; others do not.	Creates the required order from the given data.
Example	Linear search.	Bubble sort.

Searching example: In [12, 5, 18], linear search checks 12, then 5, then 18 to find 18.

Sorting example: Bubble sort arranges [12, 5, 18] as [5, 12, 18].

Q.4 OR - Describe five effects of cyberbullying. (5 marks)

1. **Emotional distress:** Victims may experience fear, sadness, anxiety, or depression.
2. **Low self-esteem:** Hurtful messages may reduce confidence and create feelings of worthlessness.
3. **Poor academic performance:** Stress may affect concentration, attendance, and examination performance.
4. **Social withdrawal:** Victims may avoid friends, school activities, or online communication.
5. **Physical health problems:** Ongoing stress may cause headaches, sleep problems, or changes in appetite.

Q.5 How does Blockchain manage cryptocurrency and asset administration? (2.5+2.5 marks)

A. Cryptocurrency

1. A user authorizes a transaction using a digital signature.
2. The network checks the transaction according to its rules.
3. Valid transactions are recorded in blocks linked to earlier blocks.
4. The shared transaction history helps establish ownership and prevent the same funds from being spent twice.

B. Asset administration

1. An asset, such as a product or ownership right, can be represented by a digital record.
2. Transfers and updates are recorded, creating a history of ownership or movement.
3. Authorized participants can check this history, helping reduce disputes and unauthorized changes.

Example: A supply-chain system can record a product's movement from its manufacturer to its retailer.

Q.5 OR - Describe website development through front-end and back-end development. (2.5+2.5 marks)

Front-end development

The front end is the visible part of a website.

1. HTML creates the structure, including headings, forms, and links.
2. CSS controls colours, fonts, spacing, and layout.
3. JavaScript adds interactive behaviour, such as responding to button clicks.

Back-end development

The back end runs on the server.

1. Languages such as Python or PHP process requests and apply business rules.
2. A database stores information such as accounts, products, and orders.
3. The back end checks information and sends results to the front end.

Example: On a login page, the front end collects the username and password. The back end checks them and returns a success or failure message.

Q.6 A school wants to analyse examination performance over several years. Which visualization should be used? Give four reasons. (1+4 marks)

Suitable visualization: Line graph. Place years on the horizontal axis and a measure such as average marks or pass percentage on the vertical axis.

1. **Shows change over time:** It clearly displays how performance changes from one year to the next.
2. **Reveals trends:** It shows whether results are improving, declining, or staying nearly the same.
3. **Allows comparisons:** Different lines can represent different subjects or classes.
4. **Highlights unusual changes:** A sharp rise or fall helps the school identify a year that needs investigation.

Q.6 OR - How do market research and customer insights help create a strong business pitch? (5 marks)

Market research collects information about the market, competitors, prices, and demand.

• It shows whether enough people need the product.

• It identifies competitors and their strengths or weaknesses.

• It supports claims about pricing and business opportunities with evidence.

Customer insights explain customers' needs, problems, preferences, and buying behaviour.

• They help define the target customers.

• They help explain how the product solves a real problem.

• They help identify the product's main benefit.

Example: If research shows that students need affordable evening tuition, a pitch can explain how an online tutoring service provides convenient lessons at a suitable price.

5. First Annual Examination 2026 - Paper M

Computer Science SSC-II | Federal Board | Complete solution with OR options

Section B - Q.2: Short Questions (22 marks)

(i) How can social media be used to understand customer preferences? Give two points. (1+1 marks)

1. Businesses can study comments, reviews, likes, and shares to identify what customers like or dislike.
2. They can use polls, surveys, and questions to collect direct feedback about products and services.

OR - How does abstraction help create generalized solutions?

Abstraction removes unnecessary details and focuses on features shared by similar problems.

This allows one solution to work in different situations. For example, a sorting function can sort the marks of any class instead of being written for only one class.

(ii) Mention two strategies that help prevent cyberbullying. (1+1 marks)

1. Teach users to communicate respectfully and avoid posting or forwarding harmful content.
2. Use privacy settings and promptly block and report accounts that send abusive messages.

OR - Why is data visualization important? Give two advantages.

1. It makes large or complex data easier to understand.
2. It helps reveal patterns, trends, and comparisons that support better decisions.

(iii) Briefly describe two real-world examples of AI applications. (2 marks)

1. **Spam filtering:** AI studies email features to identify unwanted messages.
2. **Medical support:** AI can analyse medical images and help doctors identify signs of disease.

OR - How does system software manage storage devices? Give two points.

1. It organizes data into files and folders using a file system.
2. It controls reading and writing operations through storage drivers and keeps track of available space.

(iv) How does responsible internet use promote data security? Give two points. (1+1 marks)

1. Avoiding suspicious links and downloads reduces the risk of malware and phishing.
2. Using strong passwords and sharing less personal information reduces unauthorized access and misuse of data.

OR - Why is Blockchain considered secure? Give two reasons.

1. **Cryptographic linking:** Blocks contain hashes that link them to earlier blocks, making unauthorized changes detectable.
2. **Distributed validation:** Multiple participants maintain and validate records according to agreed rules, making secret changes more difficult.

(v) Briefly explain Machine Learning in simple words. (2 marks)

Machine Learning is a branch of AI that allows computers to learn patterns from data and use them to make predictions or decisions.

Example: A spam filter learns from examples of spam and genuine emails to classify new messages.

OR - Why are encoding schemes necessary to represent text and data? Give two reasons.

1. Computers handle binary data, so encoding provides codes to represent characters and other information.
2. Standard encoding allows information to be stored and exchanged consistently between different systems.

(vi) What role does system software play during data processing by the CPU? (2 marks)

The operating system schedules programs and gives them CPU time and memory.

It also coordinates input and output so the CPU receives the required data and processing results reach the correct application or device.

OR - Differentiate between searching and sorting algorithms. Give two points.

Searching	Sorting
Finds a particular item in data.	Arranges data in an order.
Example: Linear search.	Example: Bubble sort.

(vii) How are results interpreted in the data science process? (2 marks)

Results are interpreted by examining patterns and findings and relating them to the original question or problem.

The analyst explains what the results mean and uses them to suggest conclusions or actions.

OR - Name one common encoding scheme and state its use.

ASCII is a common encoding scheme.

It represents characters such as English letters, digits, and punctuation using numerical codes, allowing computers to store and process text.

(viii) State two key differences between front-end and back-end development. (1+1 marks)

Front end	Back end
Creates the interface users see and interact with in the browser.	Handles server-side processing, rules, and database access.
Common technologies: HTML, CSS, and JavaScript.	Common technologies: Python and PHP.

OR - How does data visualization help users understand complex data?

It converts complex figures into charts, graphs, or maps.

These visuals make comparisons, patterns, and unusual values easier to recognize without reading every individual number.

(ix) Which system software controls the overall working of a computer? Give three examples. (2 marks)

The operating system controls the overall working of a computer.

Examples: Windows, Linux, and macOS.

OR - How can AI positively affect society? Give two reasons.

1. **Better healthcare:** AI can support doctors by helping analyse medical data and identify possible diseases.
2. **Greater efficiency:** AI can automate repetitive work, saving time and allowing people to focus on other tasks.

(x) Why are digital tools important for communicating ideas? Give two reasons. (1+1 marks)

1. They allow information to reach people quickly, including audiences in different locations.
2. They combine text, images, audio, and video to make ideas clearer and more engaging.

OR - Differentiate between supervised and unsupervised learning. Give two points.

Supervised learning	Unsupervised learning
Learns from labelled data, which includes correct answers.	Learns from unlabelled data, which has no supplied correct answers.
Used to predict outputs, such as spam or not spam.	Used to discover patterns or groups, such as customer groups.

(xi) What is an IDE? Enlist two modules. (1+1 marks)

An Integrated Development Environment (IDE) combines tools for developing programs in one application.

Two modules are:

1. Source-code editor.
2. Debugger.

OR - Briefly describe the pigeonhole principle. Give an example.

The pigeonhole principle states that if more objects are placed into fewer boxes, at least one box must contain more than one object.

Example: If 13 students are grouped by their 12 birth months, at least two students must have birthdays in the same month.

Section C - Long Questions (20 marks)

Q.3 How does data visualization help analysts understand patterns, trends, and relationships in a large dataset? Give five points. (5 marks)

1. **Simplifies large datasets:** Charts summarize many figures, making the overall picture easier to understand.
2. **Reveals repeated patterns:** Analysts can identify behaviour that happens regularly, such as increased shopping during holidays.
3. **Shows trends over time:** A line graph can show whether sales or expenses are rising or falling.
4. **Shows relationships:** A scatter plot can help identify a relationship between two variables, such as study hours and examination marks.
5. **Reveals unusual observations:** An unusually high or low value can be spotted and investigated for an error or an important event.

Q.3 OR - Explain responsible communication on social media, highlighting accuracy and audience awareness. (2.5+2.5 marks)

1. **Check accuracy:** Verify facts using reliable sources before posting. Incorrect information can mislead people.
2. **Give proper credit:** Mention the source of borrowed information and respect copyright.
3. **Understand the audience:** Use language and examples suitable for the audience's age, knowledge, and needs.
4. **Use clear and respectful language:** Avoid insults, hateful content, and confusing statements. Explain ideas politely.
5. **Protect privacy:** Do not share someone's private information or photographs without permission. Correct mistakes if inaccurate information has been posted.

Q.4 Evaluate the role of schools and parents in preventing cyberbullying. Suggest five measures. (5 marks)

Schools and parents can work together to teach safe behaviour, identify problems early, and support affected students.

1. **Provide awareness:** Schools should teach digital citizenship, while parents should discuss respectful online behaviour at home.
2. **Set clear rules:** Schools should have an anti-cyberbullying policy. Parents should set reasonable rules for online communication.
3. **Encourage reporting:** Students should know whom to contact and should feel safe reporting abuse without being blamed.
4. **Maintain supportive supervision:** Parents should discuss children's online experiences, and schools should supervise school-managed digital platforms.
5. **Respond promptly:** Save evidence, report harmful accounts, support the victim, and take appropriate action under school rules.

Q.4 OR - Explain CSS and JavaScript in front-end development. How do they work together? (2+2+1 marks)

CSS controls a web page's appearance. It sets colours, fonts, spacing, and layout.

Example: CSS can make a button blue and centre it on the page.

JavaScript adds behaviour and interactivity. It responds to actions such as clicking buttons or entering information.

Example: JavaScript can check whether a form field is empty.

Working together: HTML provides the structure, CSS styles it, and JavaScript controls its behaviour. For example, CSS defines an error message's red colour, while JavaScript displays that message when a user submits an incomplete form.

Q.5 How can IoT and Blockchain work together to improve security? Explain with an example. (5 marks)

IoT devices collect and exchange information using sensors and network connections. Blockchain can provide a shared, tamper-evident history of selected device records.

1. An IoT sensor collects data.
2. An authorized device or gateway submits a signed record.

3. The Blockchain network validates and records it.
4. Linked records make later unauthorized changes detectable.
5. Authorized users can check the record history and trace events.

Example: Temperature sensors monitor medicines during transport. Their readings are recorded through a Blockchain-based system. A hospital can check whether temperatures stayed within the required range and whether records were changed.

The sensors must still be secured because Blockchain cannot correct false readings supplied by a faulty or compromised sensor.

Q.5 OR - Describe five properties of a good algorithm. (5 marks)

1. **Clearly defined input:** The algorithm states what data it requires. Some algorithms may require no input.
2. **Clearly defined output:** It produces at least one result.
3. **Definiteness:** Every instruction is clear and has only one meaning.
4. **Finiteness:** It finishes after a limited number of steps.
5. **Effectiveness:** Each step is practical and can be carried out in a finite amount of time.

Q.6 Explain the importance of code repositories. How do they help version control and teamwork? (2+3 marks)

A code repository stores source code and related project files. When used with a version-control system, it provides an organized history of development.

Version control

1. **Records changes:** Each saved version, called a commit, records changes to the project.
2. **Allows recovery:** Developers can compare versions and restore earlier work when needed.

Teamwork

3. **Supports parallel work:** Developers can use separate branches to work on different features.
4. **Combines contributions:** Changes can be reviewed and merged into the shared project.
5. **Improves coordination:** Team members can track contributions and discuss proposed changes.

Example: A team can use a Git repository hosted on GitHub to develop a school website.

Q.6 OR - Design a brief elevator pitch for a digital product, including its value proposition and expected benefits. (5 marks)

Product: StudyMate

"StudyMate is a learning application for Class 9 and 10 students who struggle to organize their examination preparation. It provides short lessons, practice quizzes, and a personal study timetable in one place.

Its main value is that students can study according to their own needs and immediately identify weak topics. With affordable access and clear explanations, StudyMate helps students save time, prepare regularly, and build confidence before examinations."